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### SYSTEM AND METHOD FOR RECONSTRUCTING DATA FROM A DEGRADED RAID VOLUME USING AN ACCELERATOR ENGINE

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#### Abstract

A system and method for reconstructing data from a degraded RAID storage device using an accelerator engine is disclosed. An article of manufacture may include a non-transitory memory having machine-readable instructions that, when executed by a processor, cause the processor to send a first command to a first storage device and a second storage device to trigger the first and second storage devices to write strip data to a memory in an accelerator engine. The instructions may also cause the processor to send a second command to the accelerator engine to perform an operation on the written strip data, the operation to reconstruct data stored on a third failed storage device. The first, second, and third storage devices may be part of a RAID volume. Further, the instructions may cause the processor to receive an output of the operation from the accelerator engine.

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## Background/Summary

### PRIORITY

[0001] The present application claims priority to India Patent Application Number 202411017420, filed on Mar. 11, 2024, wherein the entire disclosure is incorporated herein by reference.

### TECHNICAL FIELD

[0002] The present disclosure relates to reconstructing data from a degraded redundant array of independent disks (RAID), e.g., where one drive of the RAID has failed, in particular, to using an accelerator engine and peer-to-peer direct memory access (DMA) capabilities to reconstruct data from a degraded RAID.

### BACKGROUND

[0003] When a group of drives is represented as a unit, i.e. the array is addressed as a unit, it may be called a RAID volume. The term RAID, while historically addressed to “disks,” may be used in relation to other storage devices, without limitation. Thus, the term RAID may be considered to refer to a redundant array of independent drives, such as solid state drives, or a redundant array of independent storage devices.

[0004] Conventional redundant array of independent disks (RAID) stacks running in hosts make use of host instructions for parity generation, which is a central processing unit (CPU) and memory intensive operation even on powerful x86\_64 servers. CPU instructions may take up to two inputs and may not be efficient for larger strips of data. Some server CPUs use advanced vector extension (AVX) instructions for parity generation which adds pressure to host and host dynamic random access memory (DRAM) for any parity calculation operations.

[0005] Exclusively-OR (XOR) parity generation is one of the building blocks of RAID algorithms. XOR parity generation is also used in various other operations like error detection, encryption, and pseudo random number generators, without limitation. Software stacks running in host servers, normally use either regular CPU instructions or advanced vector instructions like AVX-256 or AVX-512 for XOR operations. Data flows that perform XOR on multiple strips of scattered data buffers, consume significant amount of host and memory controller bandwidth to perform this XOR operation. Traditional hardware (HW) RAID architecture may be a bottleneck for performance when scaling via high-performance nonvolatile memory express (NVMe) drives.

[0006] Peripheral Component Interconnect Express (PCIe) is a high-speed serial computer expansion bus standard that replaces the older PCI, PCI-X, and Accelerated Graphics Port (AGP) bus standards. A PCIe root complex connects a PCIe end point with the CPU and memory subsystems to route communications between the connected devices. A PCIe end point or switch does not connect with the CPU or memory, but rather connects through the PCIe root complex. PCIe uses point-to-point topology, allowing for faster communication between devices.

Motherboards and systems that support PCIe, use PCIe devices of different sizes, such as  $\times 1$ ,  $\times 4$ ,  $\times 8$ , or  $\times 16$ , which refers to the number of lanes they use. PCIe devices connect to the motherboard or system using a PCIe slot so the device may be recognized by the motherboard or system.

[0007] Non-volatile memory express (NVMe) is an open, logical-device interface specification for accessing a computer's non-volatile storage media usually attached via the PCIe bus. NVMe may be used with NAND flash memory that comes in PCIe add-in cards.

[0008] However, existing RAID stacks uses the DRAM in the host for transfer buffers and the host for computations used to reconstruct a degraded RAID storage device. These computations use

significant bandwidth of the host's memory and impair the host's ability to execute other applications.

## SUMMARY OF THE INVENTION

[0009] Aspects provide systems and methods for system for reconstructing data from a degraded redundant array of independent disks (RAID) volume using an accelerator engine. An example may include an article of manufacture. The article of manufacture may include a non-transitory memory having machine-readable instructions that, when executed by a processor, cause the processor to send a first command to a first storage device and a second storage device to trigger the first and second storage devices to write strip data to a memory in an accelerator engine. The instructions may also cause the processor to send a second command to the accelerator engine to perform an operation on the written strip data, the operation to reconstruct data stored on a third failed storage device. The first, second, and third storage devices may be part of a RAID volume. Further, the instructions may cause the processor to receive an output of the operation from the accelerator engine.

[0010] In combination with any of the above examples, the instruction may further cause the processor to send the first command, send the second command, and receive the output using a Peripheral Component Interconnect Express (PCIe) bus including a PCIe root complex.

[0011] In combination with any of the above examples, the operation may be an XOR operation on the written strip data.

[0012] In combination with any of the above examples, the first and second storage devices and the accelerator engine may communicate using peer-to-peer direct memory access.

[0013] In combination with any of the above examples, the accelerator engine and the first and second storage devices may be PCIe end points.

[0014] In combination with any of the above examples, the instruction may further cause the processor to receive a message from the first and second storage devices indicating that the first and second storage devices have finished writing strip data to the memory in the accelerator engine. The message may trigger the processor to send the second command.

[0015] Alone or in combination with any of the above examples, examples of the present disclosure may include a method. The method may include sending a first command to a first storage device and a second storage device to trigger the first and second storage devices to write strip data to a memory in an accelerator engine. The first and second storage devices may be part of a RAID volume. The method may additionally include sending a second command to the accelerator engine to perform an operation on the written strip data. The operation may be to reconstruct data stored on a third failed storage device of the RAID volume. The method may further include receiving an output of the operation from the accelerator engine.

[0016] In combination with any of the above examples, the method may further include sending the first command, sending the second command, and receiving the output occurs using a Peripheral Component Interconnect Express (PCIe) bus including a PCIe root complex.

[0017] In combination with any of the above examples, the PCIe root complex may be to route communications from the first and second storage devices and the memory in the accelerator engine based on a base address register (BAR) of the memory in the accelerator engine.

[0018] In combination with any of the above examples, the operation may be an XOR operation on the written strip data.

[0019] In combination with any of the above examples, the first command may instruct the first and second storage devices to communicate with the accelerator engine using peer-to-peer direct memory access.

[0020] In combination with any of the above examples, the accelerator engine and the first and second storage devices may be PCIe end points.

[0021] In combination with any of the above examples, the method may further include receiving a message from the first and second storage devices indicating that the first and second storage

devices have finished writing strip data to the memory in the accelerator engine. The message may trigger sending of the second command.

[0022] Alone or in combination with any of the above examples, examples of the present disclosure may include a system. The system may include a memory bus, an accelerator engine circuit coupled to the memory bus, and a host coupled to the memory bus. The host may include a processor and a non-transitory memory including machine-readable instructions that, when executed by the processor, cause the processor to send a first command to a first storage device and a second storage device coupled to the memory bus to trigger the first and second storage devices to write strip data to a memory in the accelerator engine circuit. The instructions may also cause the processor to send a second command to the accelerator engine circuit to perform an operation on the written strip data. The operation may be to reconstruct data previously stored on a third failed storage. The first, second, and third storage devices may be part of a RAID volume. The instructions may further cause the processor to receive an output of the operation from the accelerator engine circuit.

[0023] In combination with any of the above examples, the memory bus may be a Peripheral Component Interconnect Express (PCIe) bus including a PCIe root complex.

[0024] In combination with any of the above examples, the PCIe root complex may be to route communications from the first and second storage devices and the memory in the accelerator engine circuit based on a base address register (BAR) of the memory in the accelerator engine circuit.

[0025] In combination with any of the above examples, the operation may be an XOR operation on the written strip data.

[0026] In combination with any of the above examples, the first and second storage devices and the accelerator engine circuit may communicate using peer-to-peer direct memory access.

[0027] In combination with any of the above examples, the accelerator engine circuit and the first and second storage devices may be PCIe end points.

[0028] In combination with any of the above examples, the instructions may further cause the processor to receive a message from the first and second storage devices indicating that the first and second storage devices have finished writing strip data to the memory in the accelerator engine circuit. The message may trigger the processor to send the second command.

[0029] Although example embodiments have been described above, other variations and embodiments may be made from this disclosure without departing from the spirit and scope of these embodiments.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0030] The figures illustrate examples of systems and methods.

[0031] FIG. 1 illustrates a block diagram of a system for reconstructing data from a degraded RAID volume using an accelerator engine, according to examples of the present disclosure;

[0032] FIG. 2 illustrates a method performed for reconstructing data from a degraded RAID volume using an accelerator engine, according to examples of the present disclosure; and

[0033] FIG. 3 illustrates a method performed for reconstructing data from a degraded RAID volume using an accelerator engine, according to examples of the present disclosure.

[0034] The reference number for any illustrated element that appears in multiple different figures has the same meaning across the multiple figures, and the mention or discussion herein of any illustrated element in the context of any particular figure also applies to each other figure, if any, in which that same illustrated element is shown.

### DESCRIPTION

[0035] According to an aspect of the invention, a system and method for reconstructing data from a degraded redundant array of independent disks (RAID) volume using an accelerator engine is provided. A RAID volume may be degraded when a storage device that is a part of the RAID volume has failed. As described above, a RAID volume is a logical device created utilizing multiple storage devices, such as drives, or disks, without limitation. The system and method reconstructs data from a failed RAID volume using peer-to-peer communications (e.g., Peripheral Component Interconnect Express (PCIe) peer-to-peer direct memory access (DMA) capability) such that an accelerator engine reconstructs data in the event of a degraded RAID volume without using memory in a host as a transfer buffer. Using the accelerator engine for data reconstruction may reduce the use of memory on a host and improve the performance of a degraded RAID volume. Memory in the accelerator engine may be used as a transfer buffer instead of memory in the host and may reduce overhead at the host.

[0036] Aspects include features disclosed in Indian Patent Application No. 202311056027 filed on Aug. 21, 2023 and U.S. patent application Ser. No. 18/527,579 filed on Dec. 4, 2023, incorporated herein in its entirety for all purposes.

[0037] FIG. 1 illustrates a block diagram of a system for reconstructing data from a degraded RAID volume using an accelerator engine, according to examples of the present disclosure. System **100** may include host **110** which may include processor **112**, memory **114**, and memory bus **116**. Processor **112** may write data to memory **114** and may read data from memory **114** using memory bus **116**. Memory **114** may be a non-transitory memory, such as non-volatile random-access memory (NVRAM), including machine-readable instructions that when executed by processor **112**, may cause processor **112** to perform one or more actions as described herein. Processor **112** may be a central processing unit (CPU), a general purpose processor, a specific purpose processor, a microcontroller, a programmable logic controller (PLC), a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, other programmable device, or any combination thereof designed to perform the functions disclosed herein. Processor **112** may drive signals onto memory bus **116** to read data from storage devices **122a**, **122b**, and **122c** (collectively “storage devices **122**”) and may drive signals onto memory bus **116** to write data to storage devices **122**. Processor **112** may include RAID driver stack **118** to manage storage devices **122** and accelerator engine **130**. Storage devices **122** may be controlled as RAID volume **120** responsive to operation of RAID driver stack **118**. As described with respect to FIG. 2, when performing data reconstruction operations (e.g., eXclusive OR (XOR) operations), RAID driver stack **118** may call a non-volatile memory express (NVMe) offload command to perform the operation. The NVMe offload command may be vendor defined. In doing so, instead of using memory **114** in host **110** for transfer buffers during data reconstruction operations, RAID driver stack **118** will use memory **132** in accelerator engine **130**.

[0038] Memory bus **116** may receive data from processor **112** and may transmit data to one or more circuits coupled to memory bus **116**. Memory bus **116** may be a Peripheral Component Interconnect Express (PCIe) bus or another bus type not specifically mentioned. Memory bus **116** may be a PCIe root complex that is the root of a hierarchy that connects with host **110**, RAID volume **120**, and accelerator engine **130**. Host **110**, storage devices **122** in RAID volume **120**, and accelerator engine **130** may be PCIe end points. A PCIe end point does not connect directly with processor **112** or memory **116**, but rather connects through memory bus **116**.

[0039] Storage devices **122** may be non-transitory storage devices, such as solid state drives or hard disk drives, including but not limited to Dynamic Random Access Memory (DRAM), Non-Volatile Memory (NVM), Embedded Non-Volatile Memory (eNVM), Non-Volatile Memory Express (NVMe), or another type of non-transitory storage not specifically mentioned. A given storage device **122** may include a circuit to move data to and from the storage device **122**. Storage devices **122** may use open, logical-device interface specifications for accessing non-volatile storage

media, wherein the specifications may include NVMe or non-volatile memory host controller interface specification (NVMHCIS). Memory bus **116** may facilitate data transmission to and from storage devices **122**. Processor **112** may move data through memory bus **116** to storage devices **122**, and processor **112** may receive data from storage devices **122** through memory bus **116**. Additionally, storage devices **122** and accelerator engine **130** may communicate with one another using memory bus **116**.

[0040] RAID volume **120** may be a redundant array of independent disks formed of storage devices **122**. In some examples, RAID volume **120** may be a RAID-5 system including a minimum of three storage devices **122**. While there is no limitation as to the maximum number of drives, for illustration purposed three storage devices will be utilized, without limitation. RAID volume **120** may use disk striping with parity when storing information. The data and parity information may be striped evenly across storage devices **122** in RAID volume **120**. As an example of disk striping, in a RAID volume with three storage devices and assuming each strip is 16 KB, each row of the RAID volume will contain two strips of data and one strip of parity information. The two strips of data are collectively referred to as a “stripe,” and will be accompanied by one strip of parity information for a given stripe. For example, when RAID volume **120** includes three storage devices **122**, a stripe may include data stored on two of the three storage devices **122** and parity information stored on the third of the three storage devices **122**. Specifically, as illustrated in FIG. 1, stripe **124a** may include data A1 and A2, stored on storage devices **122a** and **122b**, respectively, and parity information P1(A1, A2) for data A1 and A2 stored on storage device **122c**. Any two of the three storage devices **122** may store the data portion of the stripe and the remaining storage device **122** may store the parity information. As shown in FIG. 1, parity information for stripe **124b** is stored on storage device **122b**, parity information for stripe **124c** is stored on storage device **122a** and parity information for stripe **124d** is stored on storage device **122b**. Striping may allow data writes and reads to occur in parallel to multiple storage devices **122** and may avoid any single storage device **122** being a bottleneck. Striping also may enable users to reconstruct data in the event one of storage devices **122** fails, i.e. RAID volume **120** is degraded. RAID volume **120** may be considered degraded if any one storage device **122** of RAID volume **120** is missing or has failed.

[0041] System **100** may also have one or more accelerator engines **130** coupled to host **110** and storage devices **122** through memory bus **116**. Accelerator engine **130** may be used to offload computational operations from host **110** to a PCIe end point using NVMe transport protocol instruction communications and DMA data communications. Accelerator engine **130** may include memory **132**, engine **136**, and processor **138**. Accelerator engine **130** may support XOR offload commands for reconstructing data stored on a failed storage device **122** of a degraded RAID volume **120**. Accelerator engine **130** supports XOR offload commands by providing memory **132** in accelerator engine **130** that may be used as a buffer during data reconstruction instead of memory **114** in host **110**. Specifically, memory **132** may include instructions to create a region of memory **132**, such as a dynamic random access memory (DRAM) window that is exposed to a RAID driver on host **110** as a base address register (BAR) (e.g., BAR4). Memory **132** may be double data rate (DDR) memory or double data rate synchronous dynamic random access memory (DDR SDRAM), without limitation. Engine **136** may be an engine or circuit to perform an operation, such as a parity generation engine, or a parity generation circuit, and may be used to reconstruct data based on parity information. Engine **136** may perform XOR operations. Processor **138** may be a general purpose processor, a specific purpose processor, a microcontroller, a programmable logic controller (PLC), a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, other programmable device, or any combination thereof designed to perform the functions disclosed herein.

[0042] When a storage device **122** in RAID volume **120** fails, resulting in RAID volume **120** being degraded, system **100** may reconstruct the data from the failed storage device **122** using peer-to-

peer direct memory access communications between the remaining storage devices **122** (e.g., the storage devices **122** that are not failed) and memory **132** in accelerator engine **130** via memory bus **116**. When memory bus **116** is a PCIe bus, data from a failed storage device **122** may be reconstructed using PCIe peer-to-peer communication which enables two PCIe devices (e.g., storage devices **122** and accelerator engine **130**) to directly transfer data between each other without using memory **116** in host **110** as a temporary storage.

[0043] For example, in an example where storage device **122b** has failed, the reconstruction of the data from storage device **122b** may begin when processor **112** writes to a doorbell register of storage devices **122a** and **122c** using memory bus **116**. Doorbell registers provide a way for one device coupled to memory bus **116** to send a message to another device coupled to memory bus **116** without using an interrupt. Therefore, when processor **112** writes to the doorbell register of storage devices **122a** and **122c**, the doorbell register indicates to a processor or control circuit of the storage devices **122a** and **122c** that a command is pending.

[0044] The doorbell register write may cause storage devices **122a** and **122c** to read a first command from host **110** using memory bus **116**. The first command may cause storage devices **122a** and **122c** to write strip data to memory **132**. The strip data written to memory **132** may be the data associated with a given data strip. For example, when the given data stripe is data stripe **124a**, storage device **122a** may write data **A1** to memory **132** and storage device **122c** may write parity information **P1(A1, A2)** to memory **132**. Storage devices **122a** and **122c** may write the strip data to memory **132** using direct memory access (DMA) over memory bus **116** to a region of memory **132** that may be exposed to storage devices **122a** and **122c** as BAR4.

[0045] After writing strip data to memory **132**, storage devices **122a** and **122c** may send a message to a RAID driver stack via memory bus **116** indicating that storage devices **122a** and **122c** have finished writing strip data. The message may be in the form of an interrupt or a poll. For example, the message may be a message signaled interrupt (MSI) signal, such as MSI or MSI-X. In response to the message indicating that storage devices **122a** and **122c** have finished writing strip data, host **110** initiates a second command, which second command is addressed to accelerator engine **130**, e.g., by performing a write to a doorbell register at accelerator engine **130**. The second command may be an XOR command. The second command may be sent from host **110** to accelerator engine **130** via memory bus **116**.

[0046] In response to the write to the doorbell register of accelerator engine **130**, accelerator engine **130** may pull the second command from host **110** via memory bus **116**. Engine **136** in accelerator engine **130** then may perform the operation specified in the second command on the strip data written to memory **132** by storage devices **122a** and **122c**. For example, the second command may be an XOR command. The result of the second command is the reconstructed data from failed storage device **122b**.

[0047] Once the accelerator engine **130** has reconstructed the data from storage device **122b**, accelerator engine **130** may send the reconstructed data to memory **114** via memory bus **116**. After sending the reconstructed data to memory **114**, accelerator engine **130** may also send a message to a RAID driver stack over memory bus **116** indicating that data reconstruction is complete. After receiving the message, the RAID driver stack may perform any further processing on the reconstructed data. The message may be in the form of a poll or an interrupt, such as an MSI or MSI-X.

[0048] The process described above may use peer-to-peer DMA operations between storage devices **122a** and **122c** and accelerator engine **130** without sending strip data to memory **114**, and without requiring processor **112** to perform the XOR, or other, operation on data to recover the lost data, thus increasing the efficiency and reducing overhead in host **110** when reconstructing data from a degraded storage device.

[0049] FIG. 2 illustrates a method performed for reconstructing data from a degraded RAID volume using an accelerator engine circuit, according to examples of the present disclosure.

Method **200** may be implemented using any of the components shown in FIG. **1**, such as host **110**, RAID volume **120**, and accelerator engine **130**, by themselves or in combination, or any other component operable to implement method **200**. Although examples have been described above, other variations and examples may be made from this disclosure without departing from the spirit and scope of these disclosed examples.

[0050] Method **200** begins at block **210** where the RAID driver stack may send a first command to the storage devices that have not failed, or causing an error, (hereinafter “functional storage devices”) to fetch strip data from the functional storage devices and instruct the functional storage devices to write strip data to a memory in an accelerator engine circuit, such as memory **132** in accelerator engine **130** shown in FIG. **1**. The RAID driver stack may send the first command after initiating a strip read to a RAID volume and discovering that a storage device of the RAID volume has failed, or errored. In response to receiving the first command, the functional storage devices may write strip data to the memory in the accelerator engine circuit.

[0051] At block **250**, the RAID driver stack may send a second command, such as an XOR command, to the accelerator engine circuit to trigger the accelerator engine circuit to perform an operation, e.g., an XOR operation without limitation, on the strip data written at block **210**. The second command may be sent utilizing a memory write transaction to a doorbell register in the accelerator engine circuit and be a vendor defined NVMe XOR command. In response to receiving the second command, the accelerator engine circuit may perform an operation, such as an XOR or other operation, on the strip data from the memory in the accelerator engine circuit.

[0052] At block **280**, the RAID driver stack may receive the output of the operation from the accelerator engine circuit. The accelerator engine circuit may write the output of the operation to memory in the host. The accelerator engine circuit may write the output of the operation to memory in the host by DMA. The output of the XOR operation may be a reconstruction of the data stored on the failed, or errored, storage device.

[0053] Although FIG. **2** discloses a particular number of operations related to method **200**, method **200** may be executed with greater or fewer operations than those depicted in FIG. **2**. In addition, although FIG. **2** discloses a certain order of operations to be taken with respect to method **200**, the operations comprising method **200** may be completed in any suitable order.

[0054] FIG. **3** illustrates a method performed for reconstructing data from a degraded RAID volume using an accelerator engine circuit, according to examples of the present disclosure.

Method **300** may be implemented using any of the components shown in FIG. **1**, such as host **110**, RAID volume **120**, and accelerator engine **130**, by themselves or in combination, or any other component operable to implement method **300**. Although examples have been described above, other variations and examples may be made from this disclosure without departing from the spirit and scope of these disclosed examples.

[0055] Method **300** begins at block **310** where the RAID driver stack may initiate a strip read of a RAID volume, such as RAID volume **120** shown in FIG. **1**, and may discover that the RAID volume is degraded because one of the constituent storage devices has failed or errored. The RAID driver stack may initiate NVMe commands to storage devices in the RAID volume over a memory bus, such as memory bus **116** shown in FIG. **1**, and write to a doorbell register in the storage devices that have not failed. For example, referring to FIG. **1**, if storage device **122b** has failed, the RAID driver stack may write to a doorbell register in storage devices **122a** and **122c**. The doorbell register may inform the storage devices that have not failed of a pending command and trigger the storage devices that have not failed to pull a command from the host to cause storage devices that have not failed, e.g., storage devices **122a** and **122c**, to write strip data to memory in the accelerator engine circuit, such as accelerator engine **130** shown in FIG. **1**, as described with respect to blocks **320** and **330**.

[0056] At block **320**, the storage devices that received the doorbell register write (at block **210**) may, at least partially responsive to the doorbell register write, pull the first command using a



memory read to memory in the host, such as memory **114** shown in FIG. **1**. The first command may instruct the storage devices to write strip data to memory in the accelerator engine circuit, as described with respect to block **330**.

[0057] At block **330**, the storage devices that received the doorbell register write may trigger direct memory access (DMA) to write strip data to memory in the accelerator engine circuit. The memory in the accelerator engine circuit may include a DDR region exposed to the storage devices exposed as a BAR.

[0058] At block **340**, the storage devices that received the doorbell register write may send a first message to the RAID driver stack indicated that the command has completed. The message may communicate to the RAID driver stack that the storage devices have completed writing the strip data to the memory in the accelerator engine circuit and the RAID driver stack may proceed with the data reconstruction process. The first message may be in the form of an interrupt, such as an MSI or MSI-X, or a poll.

[0059] At block **350**, the RAID driver stack may initiate a second command, at least partially in response to receiving the first message (at block **340**), addressed to the accelerator engine circuit, by writing to a doorbell register in the accelerator engine circuit. The second command may be an XOR command and may be a vendor defined NVMe command.

[0060] At block **360**, in response to the doorbell register write (block **350**), the accelerator engine circuit may pull the second command from the host. The second command may be in memory at the host or any BAR.

[0061] At block **370**, an engine, or circuit, in the accelerator engine circuit may, at least partially responsive to the second command, perform an operation on the strip data from the memory in the accelerator engine circuit. The operation may be an XOR operation. The strip data used for the XOR operation may be the data written to the memory in the accelerator engine circuit at block **330**.

[0062] At block **380**, the accelerator engine circuit may send the output of the operation (from block **370**) to memory in the host. The accelerator engine circuit may write the output of the operation to memory in the host by DMA. The output of the operation may be a reconstruction of the data stored on the failed, or errored, storage device.

[0063] At block **390**, the accelerator engine circuit may send a second message to the host indicating that the command has been completed. The second message may be in the form of an interrupt, such as an MSI or MSI-X, or a poll.

[0064] Although FIG. **3** discloses a particular number of operations related to method **300**, method **300** may be executed with greater or fewer operations than those depicted in FIG. **3**. In addition, although FIG. **3** discloses a certain order of operations to be taken with respect to method **300**, the operations comprising method **300** may be completed in any suitable order.

[0065] Although examples have been described above, other variations and examples may be made from this disclosure without departing from the spirit and scope of these disclosed examples.

## Claims

**1.** An article of manufacture comprising: a non-transitory memory including machine-readable instructions that, when executed by a processor, cause the processor to: send a first command to a first storage device and a second storage device to trigger the first and second storage devices to write strip data to a memory in an accelerator engine; send a second command to the accelerator engine to perform an operation on the written strip data, the operation to reconstruct data stored on a third failed storage device, wherein the first, second, and third storage devices are part of a RAID volume; and receive an output of the operation from the accelerator engine.

**2.** The article of manufacture of claim 1, wherein the processor is to send the first command, send the second command, and receive the output using a Peripheral Component Interconnect Express

(PCIe) bus including a PCIe root complex.

3. The article of manufacture of claim 1, wherein the operation is an XOR operation on the written strip data.

4. The article of manufacture of claim 1, wherein the first and second storage devices and the accelerator engine communicate using peer-to-peer direct memory access.

5. The article of manufacture of claim 4, wherein the accelerator engine and the first and second storage devices are PCIe end points.

6. The article of manufacture of claim 1, wherein the processor is to receive a message from the first and second storage devices indicating that the first and second storage devices have finished writing strip data to the memory in the accelerator engine; and wherein the message triggers the processor to send the second command.

7. A method, comprising: sending a first command to a first storage device and a second storage device to trigger the first and second storage devices to write strip data to a memory in an accelerator engine, wherein the first and second storage devices are part of a RAID volume; sending a second command to the accelerator engine to perform an operation on the written strip data, the operation to reconstruct data stored on a third failed storage device of the RAID volume; and receiving an output of the operation from the accelerator engine.

8. The method of claim 7, wherein sending the first command, sending the second command, and receiving the output occurs using a Peripheral Component Interconnect Express (PCIe) bus including a PCIe root complex.

9. The method of claim 8, wherein the PCIe root complex is to route communications from the first and second storage devices and the memory in the accelerator engine based on a base address register (BAR) of the memory in the accelerator engine.

10. The method of claim 7, wherein the operation is an XOR operation on the written strip data.

11. The method of claim 7, wherein the first command instructs the first and second storage devices to communicate with the accelerator engine using peer-to-peer direct memory access.

12. The method of claim 8, wherein the accelerator engine and the first and second storage devices are PCIe end points.

13. The method of claim 7, comprising receiving a message from the first and second storage devices indicating that the first and second storage devices have finished writing strip data to the memory in the accelerator engine; and wherein the message triggers sending of the second command.

14. A system, comprising: a memory bus; an accelerator engine circuit coupled to the memory bus; and a host coupled to the memory bus, the host including a processor and a non-transitory memory including machine-readable instructions that, when executed by the processor, cause the processor to: send a first command to a first storage device and a second storage device coupled to the memory bus to trigger the first and second storage devices to write strip data to a memory in the accelerator engine circuit; send a second command to the accelerator engine circuit to perform an operation on the written strip data, the operation to reconstruct data previously stored on a third failed storage, wherein the first, second, and third storage devices are part of a RAID volume; and receive an output of the operation from the accelerator engine circuit.

15. The system of claim 14, wherein the memory bus is a Peripheral Component Interconnect Express (PCIe) bus including a PCIe root complex.

16. The system of claim 15, wherein the PCIe root complex is to route communications from the first and second storage devices and the memory in the accelerator engine circuit based on a base address register (BAR) of the memory in the accelerator engine circuit.

17. The system of claim 14, wherein the operation is an XOR operation on the written strip data.

18. The system of claim 14, wherein the first and second storage devices and the accelerator engine circuit communicate using peer-to-peer direct memory access.

19. The system of claim 15, wherein the accelerator engine circuit and the first and second storage

devices are PCIe end points.

**20.** The system of claim 14, wherein the processor is to receive a message from the first and second storage devices indicating that the first and second storage devices have finished writing strip data to the memory in the accelerator engine circuit; and wherein the message triggers the processor to send the second command.

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